

Section 7.4 Exercise Answers

1. $\theta = \frac{5\pi}{4}, \frac{7\pi}{4}$
2. $\theta = \frac{\pi}{3}, \frac{2\pi}{3}$
3. $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$
4. $\theta = \frac{3\pi}{4}, \frac{5\pi}{4}$
5. $\theta = \frac{3\pi}{4}, \frac{7\pi}{4}$
6. $x = \frac{\pi}{4}, \frac{5\pi}{4}$
7. $x = \frac{3\pi}{4}, \frac{7\pi}{4}$
8. $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$
9. $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$
10. $\theta = \frac{\pi}{4}, \frac{7\pi}{4}$
11. $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$
12. $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$
13. $\theta = \frac{4\pi}{3}, \frac{5\pi}{3}$
14. $\theta = \frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}, \frac{25\pi}{18}, \frac{29\pi}{18}$
15. $\theta = \frac{\pi}{3}, \frac{4\pi}{3}, \frac{\pi}{6}, \frac{7\pi}{6}$
16. $\theta = \frac{\pi}{4}, \frac{5\pi}{12}, \frac{11\pi}{12}, \frac{13\pi}{12}, \frac{19\pi}{12}, \frac{7\pi}{4}$
17. $\theta = \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{17\pi}{12}, \frac{19\pi}{12}$
18. $\theta = \frac{1}{6}, \frac{5}{6}, \frac{13}{6}, \frac{17}{6}, \frac{25}{6}, \frac{29}{6}, \frac{37}{6}$
19. $\theta = \frac{5}{6}$
20. $x = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}$
21. $x = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi$
22. $t = \frac{\pi}{3}, \pi, \frac{5\pi}{3}$
23. $t = \frac{\pi}{3}, \frac{5\pi}{3}$
24. $x = \frac{\pi}{3}, \frac{3\pi}{2}, \frac{5\pi}{3}$
25. $\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$
26. $x = 0, \pi$
27. $x = \frac{3\pi}{4}, \frac{7\pi}{4}$
28. $x = 0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4}, \frac{7\pi}{4}$
29. $t = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$
30. $t = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$
31. $x = 0, \frac{\pi}{3}, \pi, \frac{4\pi}{3}$
32. $x = \frac{\pi}{2}$
33. $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

34. No solutions

35. $x = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}$

36. $x = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$

37. $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

38. $t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$